

Usefulness of serum adenosine deaminase for diagnosing pulmonary and extrapulmonary tuberculosis

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Background

Serum adenosine deaminase (ADA) measurement is a biochemical test that is believed to play a role in diagnosing tuberculosis (TB). However, few studies have been conducted in this area and there have been contradictory reports on the diagnostic value of serum ADA in TB.

Objective

The aim of the study was to evaluate the usefulness of ADA measurement in the serum for diagnosing TB.

Materials and methods

In all, 97 participants were included in this study; 28 with pulmonary tuberculosis (PTB), 20 with extrapulmonary tuberculosis (EPTB), 32 with non-TB lung disease, and 17 healthy individuals. Serum ADA activity was determined in all participants using the semiautomated biochemistry analyzer ERMA AE600N (ERMA Inc.).

Results

We found that serum ADA levels were significantly high in patients with TB (PTB and EPTB) compared with non-TB lung disease cases and healthy control. The best cutoff value in this study of serum ADA in diagnosing TB was more than 21.1 IU/l for PTB and more than 21.2 IU/l for EPTB as obtained by the receiver operating characteristic curve with high sensitivity and specificity.

Conclusion

According to the results indicating high sensitivity for serum ADA level, this test is a helpful tool for the diagnosis of TB. On the basis of high specificity for serum ADA level, this test is also a useful test to rule out TB in questionable cases.

Keywords:

Assiut, serum adenosine deaminase, tuberculosis

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Introduction

Tuberculosis (TB) is a problem of global importance; it is a medical, social, and economic disaster of immense magnitude that is occurring over the world [1,2]. Ten million TB cases have been reported in 2017, and about 1.57 million cases have died [3,4]. In Egypt, researchers have found evidence of TB in Egyptian mummies dating back to 3000–2400 BC [5] and WHO stated that TB is the most important public health problem after hepatitis C [6]. TB disease can affect any organ, but it commonly occurs in the lungs which is called pulmonary TB (PTB). The term extrapulmonary tuberculosis (EPTB) refers to isolated TB anywhere in the body outside the lungs [7]. According to WHO, the top priority for effective TB control is the rapid diagnosis and treatment of the disease. However, diagnosis of TB can be easily missed as TB usually shows unspecified and varied signs and symptoms [8]. Several methods are used to diagnose TB and several attempts have been made to improve the techniques for detecting the acid-fast bacilli (AFB) or its components. However, these methods have limitations in terms of sensitivity, cost, or availability [9]. Additionally, the procedures used to obtain a

sample to diagnose EPTB are invasive, risky, and not always accepted by patients [10]. Hence, simple, fast, and reliable testing is needed that can be easily performed in a clinical laboratory. Accordingly, there is great interest in developing tests based on the biochemical response of the body toward TB infection. Adenosine deaminase (ADA) is an enzyme of purine catabolism that is abundant in lymphocytes responsible for the adaptive immune response to control TB infection, and eventually ADA activity in TB increases [9]. Serum ADA measurement is a biochemical test that is believed to play a role in diagnosing TB. However, few studies have been conducted in this area and there have been contradictory reports on the diagnostic value of serum ADA in TB [11,12]. The aim of our study was to evaluate the usefulness of ADA measurement in the serum for diagnosing PTB and EPTB.

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Materials and methods

This study was carried out as a comparative cross-sectional study in Assiut University Hospital, which is affiliated to the Faculty of Medicine at Assiut Governorate, located in Upper Egypt. Between May 2019 and December 2019, a total of 97 participants (age 18 years and older) were enrolled in this study and were divided to four groups. The first group [pulmonary tuberculosis (PTB) group] consisted of 28 patients with PTB diagnosed based on the detection of the AFB in the respiratory specimen. The second group (EPTB group) consisted of 20 patients with EPTB diagnosed based on the detection of *Mycobacterium tuberculosis* in specimens obtained from the lesion or proved TB granuloma in the tissue affected. Excluded from group A and group B are patients who were on anti-TB therapy. The third group [nontuberculosis lung diseases (NTLD) group] consisted of 32 patients including 10 patients with chronic obstructive pulmonary disease (COPD), 10 with bacterial pneumonia, five with bronchiectasis, and seven with lung cancer. Patients of this group were included provided associated TB was excluded. The fourth group (control group) consisted of 17 volunteers who were healthy in every aspect. Excluded from this group are those who had a past history of TB, family history of TB, and history of close contact with TB patients. In general, it was excluded from this study patients who had immunocompromised status, liver diseases, kidney diseases, diabetes mellitus, lymphoma, and those on systemic steroids. Our study was accepted by the Scientific Ethics Committee of Faculty of Medicine of Assiut University. Before enrollment, informed written consent was obtained from each participant and we confirmed that privacy is maintained by not publishing the identifying data. Demographic and clinical data were collected and serum ADA activity was determined in all participants. Using the aseptic technique, a 3 ml venous blood sample was taken and tested for ADA using the semiautomated biochemistry analyzer ERMA AE600N (ERMA Inc., Tokyo, Japan); the results were determined by the GLDH/UV kinetic method on spectrophotometer and interpretation of test results was as per manufacturer's guidelines. Serum ADA activity was expressed in IU/l.

Statistical analysis

Data were collected and analyzed using SPSS (Statistical Package for the Social Sciences, version 20; IBM, Armonk, New York, USA). Continuous data were expressed in the form of mean $\pm$ SD or median (range) while nominal data were expressed in the form of frequency (percentage). χ^2 -test was

used to compare the nominal data of different groups in the study while analysis of variance test was used to compare the mean of continuous variables between different studied groups followed by post-hoc analysis. The receiver operating characteristic curve was used to predict the diagnostic accuracy of ADA in the diagnosis of PTB and EPTB. The level of confidence was kept at 95% and hence the *P* value was significant if less than 0.05.

Results

Ninety-seven participants were included in the current study; 28 (28.9%) with PTB, 20 (20.6%) with EPTB, 32 (33%) with NTLD, and 17 (17.5%) healthy individuals (the control group) (Fig. 1).

The majority of participants among the different studied groups were men. There was significant differences between different groups as regards age ($P=0.04$). With post-hoc analysis, there was significant differences between EPTB and NTLD as regards age (38.50 ± 15.14 vs 47.18 ± 11.64 years, $P=0.01$) and between PTB and NTLD (39.53 ± 12.94 vs 47.18 ± 11.64 years; $P=0.02$) (Table 1).

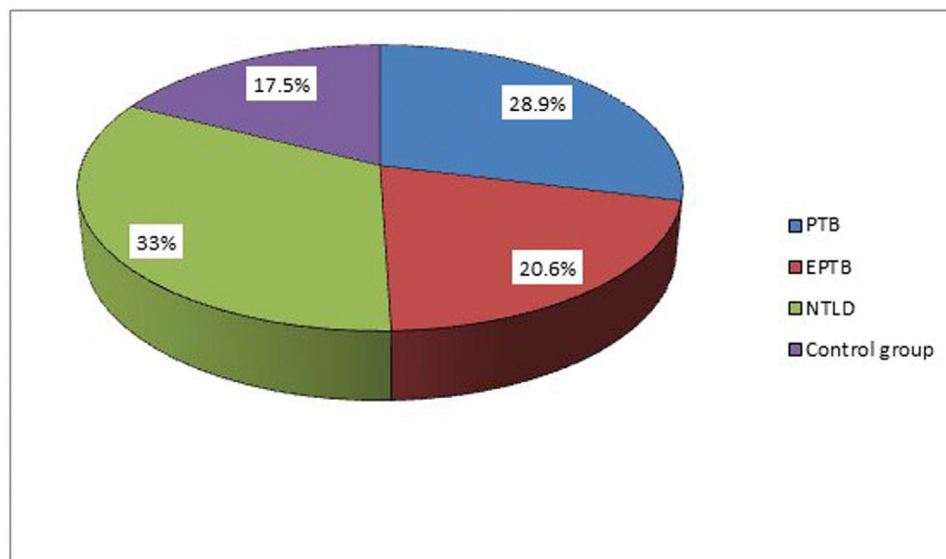
There was a significant difference among the studied groups as regards the serum level of ADA ($P=0.02$). With post-hoc analysis, it was noticed that the level of ADA had significant differences between different groups with each other except for the difference between PTB and EPTB (32.61 ± 10.68 vs 31.45 ± 12.32 ; $P=0.65$) on one hand and between NTLD and control group (19.33 ± 6.30 vs 14.55 ± 1.94 ; $P=0.07$) on the other hand (Table 2).

Regarding the diagnostic performance of serum ADA in TB, at a cutoff point of greater than 21.1 IU/l, serum ADA had 89.3% sensitivity and 84% specificity for the diagnosis of PTB with overall diagnostic accuracy was 87%, while at a cutoff point greater than 21.2 IU/l; ADA had 94% sensitivity and 83% specificity for the diagnosis of EPTB with overall diagnostic accuracy of 90.4% (Table 3, Figs 2 and 3).

Discussion

First, before we get into discussing our findings, it is worth noting that this study is the first of its kind in Assiut University Hospital to evaluate the role of serum ADA in diagnosing TB. Second, certain circumstances, such as the negative results of the available tests for AFB detection and the high cost of conducting some investigations, in addition to the difficulty of obtaining the appropriate samples required for diagnosis may sometimes lead to a decision to start the antituberculous treatment empirically

Figure 1



Different studied groups in this study.

Table 1 Age and sex distribution among the studied groups

	PTB (n=28)		EPTB (n=20)	NTLD (n=32)	Control (n=17)	P
Age (years)	39.53±12.94		38.50±15.14	47.18±11.64	40.05±1.98	0.04
Sex						0.69
Male	16 (57.1)		11 (55.6)	19 (59.4)	9 (52.9)	
Female	12 (42.9)		9 (45)	13 (40.6)	8 (47.1)	
Significance	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆
Age	0.78	0.02	0.89	0.02	0.71	0.06
Sex	0.21	0.09	0.19	0.08	0.07	0.10

Data expressed as frequency (percentage) and mean±SD. EPTB, extrapulmonary tuberculosis; NTLD, nontuberculous lung disease; P_1 compared between PTB and EPTB; P_2 compared between PTB and NTLD; P_3 compared between PTB and control group; P_4 compared between EPTB and NTLD; P_5 compared between EPTB and control; P_6 compared between NTLD and control; PTB, pulmonary tuberculosis. P value was significant if <0.05.

Table 2 Level of serum adenosine deaminase among the studied groups

	PTB (n=28)	EPTB (n=20)	NTLD (n=32)	Control (n=17)	P	
ADA (IU/l)					0.01	
Mean±SD	32.61±10.68	31.45±12.32	19.33±6.30	14.55±1.94		
Range	13.80–55.50	20.10–70.40	12.30–37	11.40–18.30		
Significance	<i>P</i> ₁	<i>P</i> ₂	<i>P</i> ₃	<i>P</i> ₄	<i>P</i> ₅	<i>P</i> ₆
	0.65	<0.001	<0.001	<0.001	<0.001	0.07

Data expressed as frequency (percentage) and mean±SD. ADA, adenosine deaminase; EPTB, extrapulmonary tuberculosis; NTLD, nontuberculous lung disease; P_1 compared between PTB and EPTB; P_2 compared between PTB and NTLD; P_3 compared between PTB and control group; P_4 compared between EPTB and NTLD; P_5 compared between EPTB and control; P_6 compared between NTLD and control; PTB, pulmonary tuberculosis. P value was significant if <0.05.

in our institution, which necessitated the idea of this study for further evaluation of a simple, available, and inexpensive laboratory test that may make such a decision based on more accurate information and reduces the possibility of error.

Regarding our main findings, the mean serum ADA level recorded in the healthy individuals was 14.5 IU/l, which approximates the levels recorded by Badade *et al.* (14.6 IU/l) [13], Alaarag *et al.* (14.9 IU/l) [14], and Ninghot *et al.*

(13.3 IU/l) [9]. However, our result was different from that reported by Verma *et al.* (16.2 IU/l) [15] and Rao *et al.* (17.4 IU/l) [16] and remarkably far from that notified by Salmanzadeh *et al.* (10.7 IU/l) [17] and Lende *et al.* (20.4 IU/l) [18]. In fact, we do not have a definite explanation for the interstudy variation as regards the serum level of ADA among the healthy people, but it may be attributed to the diversity of the studied age groups, gender, and race. Additionally, the difference in the criteria on which these healthy participants were chosen in these studies may have

a role in this variation. We believe and hope this note is an area for future research.

In our study, the mean serum ADA in the studied groups (PTB, EPTB, NTLD, and healthy control) was 32.6, 31.4, 19.3, and 14.5 IU/l, respectively. We found that serum ADA levels were significantly high in patients with TB (PTB and EPTB) compared with NTLD cases and the healthy control. However, there was no significant difference between PTB and EPTB cases. The statistical significance of our results was similar to that observed in previous studies conducted by Verma *et al.* [15], Rao *et al.* [16], Hassanein *et al.* [19], Alaarag *et al.* [14], Ninghot *et al.* [9] yet, the mean serum ADA levels for patients in our study was lower than that reported by the above-mentioned studies, which we cannot explain exactly but may be due to the degree of severity of the disease. On the other hand, some studies including a review study by

Dinnes *et al.* [20] did not demonstrate significant differences in serum ADA levels, between patients with TB and other patients with non-TB infection.

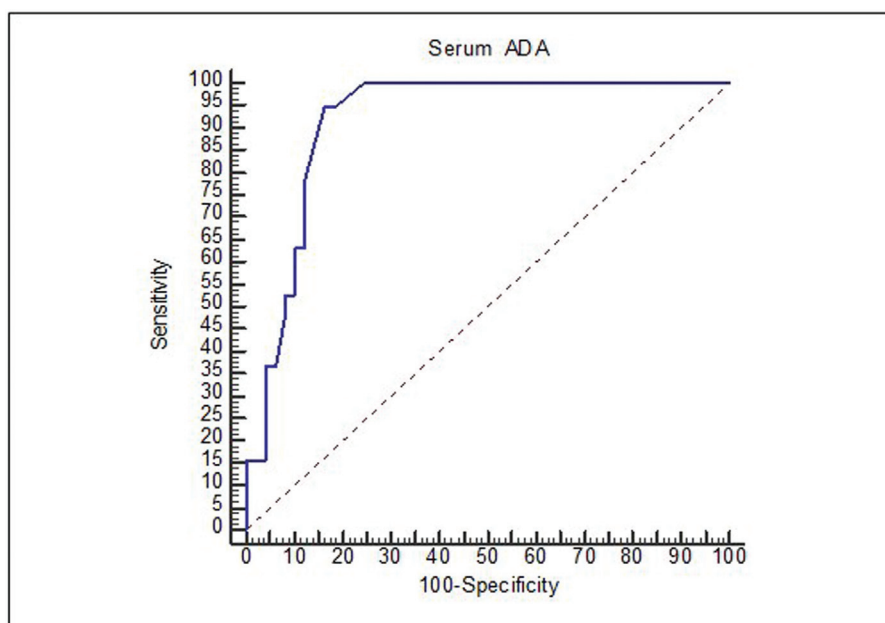
The best cutoff value in the current study of serum ADA in diagnosing TB was more than 21.1 IU/l for PTB and greater than 21.2 IU/l for EPTB as obtained by receiver operating characteristic curve with high sensitivity and specificity. Our finding related to the validity of the serum ADA test is consistent with the studies with high sensitivity as reported by Kuyucu *et al.* and Mishra *et al.* [21,22], but disagrees with studies with low sensitivity as reported by Agarwal *et al.* [23], Titarenko *et al.* [24], Kartaloglu *et al.* [25], Farazi *et al.* [12], and Salmanzadeh *et al.* [17]. Therefore, according to our results indicating the high sensitivity for serum ADA level, this test is a helpful tool for TB diagnosis. Based on the high specificity for serum ADA level, this test is also a useful test to rule out TB in questionable cases. However, what we can consider as limitation in our study is that we did not consider the age match between tuberculous and nontuberculous patients that may have contributed to the difference of the results with respect to the serum level of ADA. The difference in conclusion between different studies around the world should be taken with interest by conducting further studies in this field, taking into account factors that could have an impact on results such as age, sex, race, as well as the degree of severity and extent of the disease. We also recommend studies on the role of the serum ADA level as a prognostic marker in patients with TB and as a monitor for response to anti-TB therapy.

Table 3 Diagnostic performance of serum adenosine deaminase in diagnosing tuberculosis

Indices	PTB (%)	EPTB (%)
Sensitivity	89.3	94
Specificity	84	83
Positive predictive value	76	69
Negative predictive value	93	97
Accuracy	87	90.4
Cutoff point	>21.1	>21.2
Area under curve	0.91	0.92
P value	<0.001	<0.001

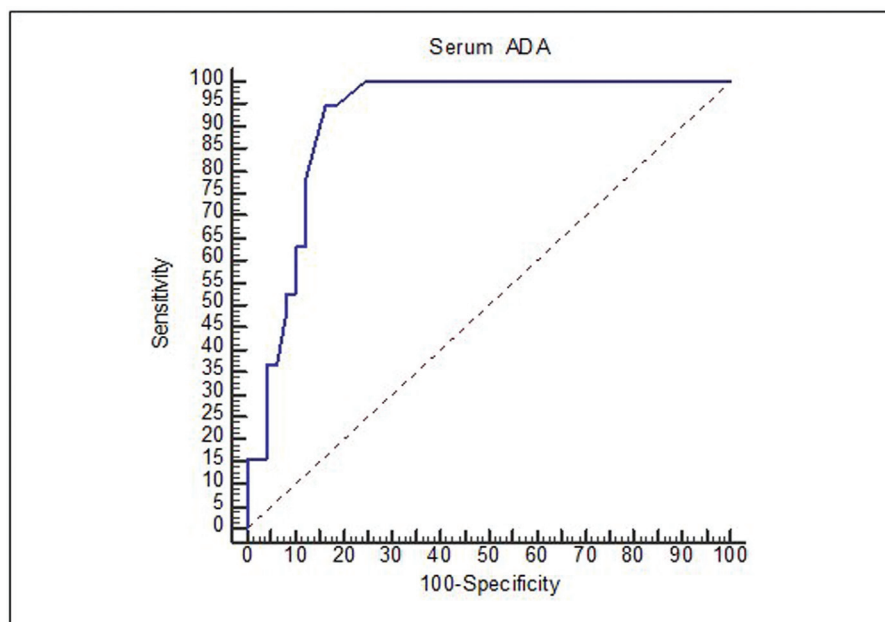
EPTB, extrapulmonary tuberculosis; PTB: pulmonary tuberculosis.

Figure 2



Accuracy of serum adenosine deaminase in the diagnosis of pulmonary tuberculosis.

Figure 3



Accuracy of serum adenosine deaminase in the diagnosis of extrapulmonary tuberculosis.

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Nil.

Conflicts of interest

There are no conflicts of interest.

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